

Emilio Luis Sáenz Guillén

Ph.D. Candidate in Actuarial Science · Bayes Business School

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SUMMARY

I am an Actuary and Economist, interested in the development of statistical methods with applications across a broad range of fields, including finance and actuarial science. I am pursuing a Ph.D. in Actuarial Science at Bayes Business School. My research to date has focused on stochastic models using spline functions and copulas, and includes work on generalised additive models, gradient boosting and non-parametric density estimation, as well as the modelling real/perceived income and standard of living, and their implications in marketing.

RESEARCH

WORKING PAPERS

- D. S. Dimitrova, V. K. Kaishev and E. Sáenz Guillén (2025). ‘GeDS: An R Package for Regression, Generalized Additive Models and Functional Gradient Boosting, based on Geometrically Designed (GeD) Splines’. Manuscript submitted for publication. Under review in the *Journal of Statistical Software*.

WORK IN PROGRESS

- ‘Density and distribution function estimation using variable-knot splines’, joint project with [Prof. Vladimir Kaishev](#) and [Dr. Dimitrina Dimitrova](#). A corresponding R package is publicly available at: [ddfs](#).
- ‘On the joint distribution of (perceived) income and standard of living, and its effect on inequality and new product diffusion’, joint project with [Prof. Vladimir Kaishev](#), [Prof. Anirban Mukhopadhyay](#) and [Dr. Jingshi Liu](#). A corresponding survey study for this project was funded by a £1,000 internal grant and implemented via [Prolific](#) and [ShinyApps](#) platforms.
- ‘Bariatric Data Analytics Based on the UK National Bariatric Surgery Registry (NBSR)’, joint project with [Prof. Vladimir Kaishev](#), [Dr. Dimitrina Dimitrova](#), and collaborators from Imperial College London, BOMSS, and the NHS.

ACADEMIC EXPERIENCE

BAYES BUSINESS SCHOOL | DOCTORAL RESEARCHER

2022–Sep/2026 (expected) | London, UK

- Research topics: splines; copulas; gradient boosting methods; non-parametric density estimation; micro-level projection of mortality; pension systems modelling.
- Supervisors: [Prof. Vladimir Kaishev](#) and [Dr. Dimitrina Dimitrova](#).

IESE BUSINESS SCHOOL | RESEARCH ASSISTANT

2021–22 | Barcelona, Spain

- Under the supervision of [Prof. Christian Eufinger](#), [Assoc. Prof. Stefano Sacchetto](#) and [Assist. Prof. Valentina Raponi](#).
- Two main projects: the first, analysed the impact of stakeholders’ ESG pressures on banking behaviour; the second, examined potential moral hazards introduced by government bailout guarantees.
- Research topics: banking and corporate finance.

IIEP UBA-CONICET | RESEARCH ASSISTANT

2019–20 | Buenos Aires, Argentina

- UBACyT grant holder under the project *Econometric methods for the selection of levels of data grouping*, led by [Prof. Gabriel Montes-Rojas](#).
- Joint work with the Central Bank of the Argentine Republic (BCRA), studying the effect of network centrality on interest rate spreads in Argentine interbank markets. Collaborated on the paper: Gabriel Montes-Rojas and Pedro Elosegui (2020). ‘Network ANOVA random effects models for node attributes’. In: *Journal of Dynamics and Games* 7.3, pp. 239–252. ISSN: 2164-6066. DOI: [10.3934/jdg.2020017](https://doi.org/10.3934/jdg.2020017). URL: <https://www.aims sciences.org/article/id/3f69030f-34ce-45a2-ba50-4a74368bdadc>.

- Volunteered with the [Índice Barrial de Precios \(IBP\)](#), a participatory research project estimating the consumer goods basket in low-income neighbourhoods of Argentina during years of unreliable official statistics. I presented a related work on wages, income, and employment in Argentina (2015–2018) at the *XI Jornadas de Economía Crítica*, organised by the [Critical Economics Society](#).

TEACHING EXPERIENCE

BAYES BUSINESS SCHOOL | TEACHING ASSISTANT

2023–Present | London, UK

- Probabilistic Modelling (AS3209) - [Prof. Vladimir Kaishev](#).
- Mathematics for Actuarial Science (AS1056) - [Dr. Russell Gerrard](#).
- Mathematics for Finance/Business Analytics (FR1202/BA1101) - [Dr. Dimitrina Dimitrova](#).

IMPERIAL COLLEGE BUSINESS SCHOOL | TEACHING ASSISTANT

2024–25 | London, UK

- Finance and Pricing Strategy (MSc Strategic Marketing) - [Prof. Mushegh Harutyunyan](#).

LONDON BUSINESS SCHOOL | TEACHING ASSISTANT

2022–24 | London, UK

- Time Series Analysis (E334; MBA programme) - [Prof. Derek Bunn](#).

UNIVERSITY OF NAVARRA | TEACHING ASSISTANT

2020–21 | Pamplona, Spain

- Marketing II and Family Business - [Assoc. Prof. Goretti Cabaleiro](#).
- Strategic Management - [Prof. Ricardo Mateo Dueñas](#).

UNIVERSITY OF BUENOS AIRES | TEACHING ASSISTANT

2020 | Buenos Aires, Argentina

- Econometrics I - [Prof. Gabriel Montes-Rojas](#).

EDUCATION

BAYES BUSINESS SCHOOL | M.PHIL./PH.D. IN ACTUARIAL SCIENCE

2022–Present | London, UK

1ST YEAR COURSES¹

London School of Economics and Political Science

Probability and Mathematical Statistics I (ST552); Probability and Mathematical Statistics II (ST553).

Bayes Business School

Probability and Mathematical Statistics (SMM941); Financial Econometrics (SMM943); Pensions and Other Benefits (SMM033; SP4); Machine Learning (SMM636); Applied Deep Learning (SMM768).

¹ Courses, both LSE and Bayes Business School, included a formal written examination and were all passed with distinction.

UNIVERSITY OF NAVARRA | MASTER IN ECONOMICS AND FINANCE

2020–21 | Pamplona, Spain

GPA: 9.28/10

Master's Thesis: 'The Chilean Pension System: Actuarial Analysis of a Paradigmatic Social Security Program'.

Supervisors: [Assist. Prof. José Azar](#) and Dr. Act. Eduardo Melinsky (Director of the Actuarial Program & Research Centre for Insurance of the University of Buenos Aires).

UNIVERSITY OF BUENOS AIRES

2013–20 | Buenos Aires, Argentina

ACTUARY DEGREE

Society of Actuaries (SOA) Recognition Tier: UCAP-AC. Level of Actuarial Courses Offered: Masters.

BACHELOR'S DEGREE IN ECONOMICS

Cum Laude distinction

Dissertation: *Application of the Solow Growth Model to Argentina (2004–2014): An Evaluation of the Impact of Capital, Labour, and Technology*. Supervisor: [Guido Lorenzo](#).

PROFESSIONAL QUALIFICATIONS AND CERTIFICATIONS

ADVANCE HIGHER EDUCATION (UK) | ASSOCIATE FELLOW (AFHEA)

May 2024

CPCE - AUTONOMOUS CITY OF BUENOS AIRES | FULLY QUALIFIED ACTUARY

Dec 2023

CONFERENCES

2025

- [28th International Congress on Insurance: Mathematics and Economics](#) held in Tartu, Estonia, on July 1-4, 2025. *Nonparametric Free-Knot Spline Density and Distribution Estimation for Heavy-Tailed Data.*
- [Insurance Data Science Conference, 19-20 June 2025](#), Bayes Business School, City St George's, University of London. *Non-Parametric Insurance Loss Modelling using Variable-Knot Splines.*
- [Perspectives on Actuarial Risks in Talks of Young Researchers 2025](#) (Liverpool, UK). *Estimation of Probability Density and Distribution Functions using Variable-Knot Splines with Applications to Insurance Loss Data.*

2024

- Bayes Business School - KU Leuven - University of Amsterdam PhD Workshop 2024. *Simultaneous Estimation of Probability Density and Distribution Functions using Variable-Knot Splines.*
- [RSS International Conference 2024](#), 2-5 September, Brighton (UK). *Augmented Spline Regression for Advanced Data Analysis: Generalized Additive Models and Functional Gradient Boosting with Geometrically Designed (GeD) Splines.*
- [26th International Conference on Computational Statistics \(COMPSTAT 2024\)](#), 27-30 August 2024, University of Giessen. *Enhancing Geometrically Designed Spline (GeDS) regression through Generalized Additive Models and Functional Gradient Boosting.*
- [Insurance Data Science Conference, 17-18 June 2024](#), Stockholm University, Department of Mathematics. *Generalized Additive Models & Functional Gradient Boosting with Geometrically Designed (GeD) Splines: Application to Insurance Data.*

OTHER WORK EXPERIENCE

GRUPO ISN | PROCUREMENT AND FINANCE MANAGER

2017–18 | Buenos Aires, Argentina

Manager of the Procurement and Finance Department during the launch of a new contract with Volkswagen. Collections | Purchases (market surveys, budget evaluations, supplier relations) | General accounting.

UNIVERSITY OF BUENOS AIRES | ADMINISTRATOR

2015–16 | Buenos Aires, Argentina

Administrative assistant at the Student Center of the Faculty of Psychology of the University of Buenos Aires. Responsibilities included managing supplier relations, processing wage payments, allocating scholarships, and organising academic seminars. Part-time position.

EL MOLINO RESTO/BAR | COOK

2014–15 | Buenos Aires, Argentina

Cook at grill restaurant. Part-time job.

SAUMA WAGEN ARGENTINA | INTERNSHIP

2012 | Buenos Aires, Argentina

100-hour accounting internship at an official Ford and Volkswagen dealership.

SUPERVISION IN ARGENTINIAN ELECTIONS

PASO and Final 2013, 2015, 2017 and 2019.

AWARDS, FELLOWSHIPS, & GRANTS

2019 | UBACyT research scholarship, University of Buenos Aires

OTHER CERTIFICATES

Introductory Certificate in Academic Practice

Learning Enhancement and Development (LEaD). City, University of London.

GRE

Quantitative reasoning: 166/170, 86th percentile.

OTHER SKILLS & EXPERIENCE

PROGRAMMING

Maintainer

• **GeDS** R Package ([CRAN](#); [GitHub](#)) • **ddfs** R Package ([GitHub](#))

Other programming & software

• Python • MATLAB • Stata • LaTeX • SQL • Bejerman Accounting System • Office (Word, Power Point, Excel)

VOLUNTEERING

2013 - 2017 Solidarity work in Villa La Cava, San Isidro (Buenos Aires, Argentina);

2011 Solidarity work in Juan Bonal Foundation (Spain).

LANGUAGES

English

Fluent. 106/120 in TOEFL; First Certificate in English.

French

Fluent. Dalf C1 Aliance Française.

German

Beginner. Goethe-Zertifikat A2: Fit in Deutsch.

Portuguese

Bilingual, 7 years living in Portugal.

Basque

Fluent.

Mother tongue

Spanish and Catalan.